Discussion 3: Vectors, Matrices, and Rotations – Solutions

Names:

Section:

This week we'll connect the mathematical concepts of vectors and matrices to physics concepts you already know, namely Newton's Laws. Our usual way of thinking of a vector is an arrow in space, with one end at the origin - the numbers (components) associated with this vector represent the endpoint of the arrow. But is there something special about our definition of a vector? Newton's laws are defined in terms of vectors - what is it about vectors that makes Newton's laws take the same form in every inertial reference frame? We will see that a more careful definition of a "vector" is closely tied to the concept of symmetry, and that stating Newton's laws as vector equations are related to deep facts about the symmetry of the universe. We will make extensive use of these concepts for the rest of this semester.

Part 1: What is a Vector?

We will work in two dimensions for this part. You need not turn in any written work, but please do perform all of the listed exercises in your group! This exercise may seem a bit silly and obvious at first, but keep these tangible concepts in mind for the rest of this discussion, because our goal now is to connect the abstract mathematical definition of a vector to these intuitive concepts.

Draw an arrow on a piece of transparency (your instructor will provide this to you) about 2 cm long, and call this vector $\mathbf{v}$.

Note: no written solutions provided for this part.

1.1. Draw an xy-plane on a second piece of transparency, and place it over the arrow in such a way that the tail of the arrow is at the origin, and the head lies on the x-axis. You have effectively defined a *coordinate system* S, one in which $\mathbf{v}$ has an x-component representing its length, and a y-component equal to zero; we could write it as an ordered pair $(L, 0)$.

1.2. Keeping the tail of the vector at the origin, rotate the top transparency so that the x-axis no longer lines up with the vector. This defines *another* system of coordinates S', in which the vector is no longer represented by $(L, 0)$, but by some other ordered pair of two numbers. *The vector itself is unchanged* – $\mathbf{v}$ is just some fixed arrow you drew – but its *representation* as a pair of coordinates differs depending on how you orient your axes. This is known as a *passive transformation* because we haven't touched $\mathbf{v}$, we've only changed our definition of the coordinate system.

1.3. Now, exchange the top and bottom transparencies and re-align them so that $\mathbf{v}$ points in the x-direction again. Rotate the top transparency, which now corresponds to *physically changing the vector* $\mathbf{v}$. This is known as an *active transformation*, after which $\mathbf{v}$ has a different set of components in the *same* frame S.

1.4. Convince yourself that in both active and passive transformations, the *length* of the vector doesn't change, no matter what coordinate system you use. If you draw a second arrow $\mathbf{w}$ on

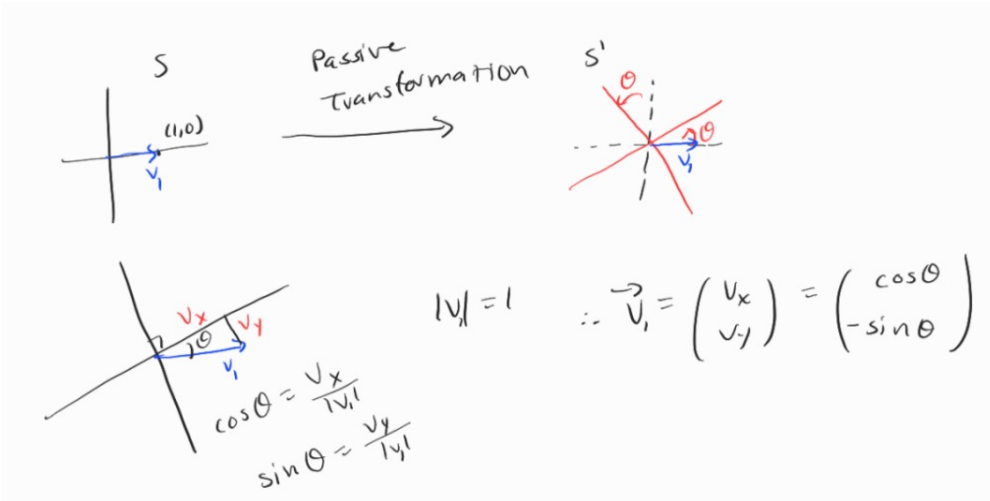

Figure 1: Passive rotation of coordinate axes by an angle θ, and the resulting relabeling of the unit vector (1,0).

the same transparency as $\mathbf{v}$, also with its tail at the origin, and you repeat the above exercises, you will also notice that the *angle* between the two vectors doesn't change no matter what coordinate system is used.

Part 2: Vectors and Rotation Matrices

To quantify the discussion in Part 1, suppose that we start with a coordinate system S (still in 2 dimensions for the time being) in which a vector $\mathbf{v}_1$ is represented by $(1, 0)$. Now we perform a passive transformation, rotating the coordinate axes counterclockwise by an angle θ.

2.1. Show that the components of $\mathbf{v}_1$ in this new coordinate system S' are $(\cos\theta, -\sin\theta)$.

We can see by means of geometric argument, depicted in Fig. 1, that when we rotate the coordinates axes counterclockwise by an angle θ, our vector $\mathbf{v}_1$ becomes $(\cos\theta, -\sin\theta)$.

2.2. Similarly, show that a second vector $\mathbf{v}_2$, represented by $(0, 1)$ in S, has components $(\sin\theta, \cos\theta)$ in the new system.

Similarly, we can see from Fig. 2 that when we rotate the coordinates axes counterclockwise by an angle θ, our vector $\mathbf{v_2}$ becomes $(\sin\theta, \cos\theta)$.

2.3. Using trig identities, prove that in both S and S', $\mathbf{v}_1$ and $\mathbf{v}_2$ have length 1 and are orthogonal to each other ($\mathbf{v}_1 \cdot \mathbf{v}_2 = 0$). This is one example of the preservation of lengths and angles mentioned in Part 1.

$$|\mathbf{v}_1| = \sqrt{\cos^2\theta + \sin^2\theta} = 1$$

$$|\mathbf{v}_2| = \sqrt{\sin^2\theta + \cos^2\theta} = 1$$

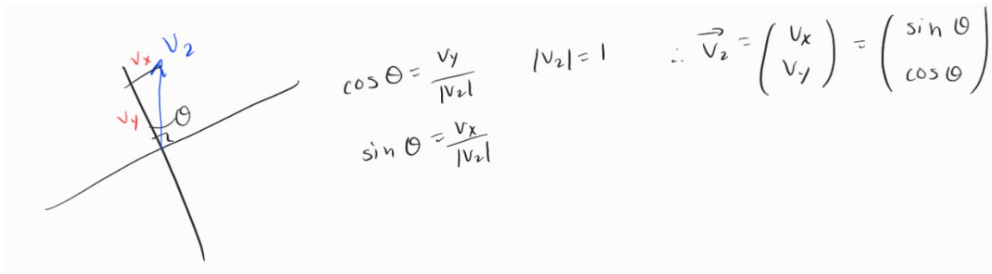

Figure 2: Passive rotation of coordinate axes by an angle θ, and the resulting relabeling of the unit vector (0,1).

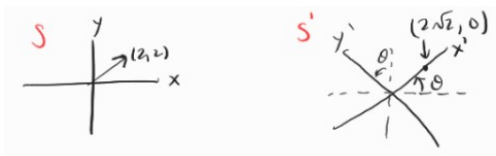

Figure 3: Passive rotation of coordinate axes by $\frac{\pi}{4}$, and the resulting relabeling of the vector (2, 2).

$$\mathbf{v}_1 \cdot \mathbf{v}_2 = \sin\theta\cos\theta - \sin\theta\cos\theta = 0$$

2.4. A basic fact from linear algebra is that defining the transformation rules for the two *basis vectors* $\mathbf{v}_1$ and $\mathbf{v}_2$ is enough to define the transformation rules for *every* vector in the xy-plane. That is, the components of any vector $\mathbf{r}$ in the two coordinate systems are related by

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} \tag{1}$$

where (x, y) are the components of $\mathbf{r}$ in S and (x', y') are the components of $\mathbf{r}$ in S'. If you haven't seen a lot of linear algebra, this might not be obvious, so check that (1) is correct by choosing a vector $\mathbf{r}$ other than $\mathbf{v}_1$ or $\mathbf{v}_2$ and computing its components in S and S'. Practicing our summation convention, we can also write this as

$$x'^i = R^i_j\, x^j \tag{2}$$

where $x^i \equiv (x, y)$, $x'^i \equiv (x', y')$, and R^i_j are the components of the 2×2 matrix R appearing in (1).

Let's select a random vector $\vec{r} = (2, 2)$ and a random angle $\theta = \frac{\pi}{4}$. Then $\mathbf{r}'$ is obtained by

$$\begin{pmatrix} r'_x \\ r'_y \end{pmatrix} = \begin{pmatrix} \cos\frac{\pi}{4} & \sin\frac{\pi}{4} \\ -\sin\frac{\pi}{4} & \cos\frac{\pi}{4} \end{pmatrix} \begin{pmatrix} 2 \\ 2 \end{pmatrix} = \begin{pmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{pmatrix} \begin{pmatrix} 2 \\ 2 \end{pmatrix} = \begin{pmatrix} 2\sqrt{2} \\ 0 \end{pmatrix}$$

We see in Fig. 3 that the coordinate axes have rotated, but we've found a proper new labeling for (2, 2).

2.5. The matrix R is the most general form of a *rotation matrix* in two dimensions. It has special properties which make it ideally suited to describing transformations of vectors. Show using (1) that $\sqrt{(x')^2+(y')^2}=\sqrt{x^2+y^2}$; that is, this matrix preserves lengths of vectors. Show also that for two vectors $\mathbf{a}$ and $\mathbf{b}$, $a'^1b'^1+a'^2b'^2=a^1b^1+a^2b^2$; that is, multiplication by R preserves dot products as well. But if the dot product and length are both preserved, then so is the angle α between $\mathbf{a}$ and $\mathbf{b}$, since $\mathbf{a}\cdot\mathbf{b}=|\mathbf{a}||\mathbf{b}|\cos\alpha$. So a rotation matrix preserves lengths and angles of arbitrary vectors - just the properties we need to satisfy our intuitive conclusions in Part 1.

$$x' = \cos\theta + \sin\theta$$

$$y' = -\sin\theta + \cos\theta$$

$$(x')^2+(y')^2 = (\cos\theta+\sin\theta)^2 + (-\sin\theta+\cos\theta)^2$$

$$(x')^2+(y')^2 = x^2\left(\cos^2\theta+\sin^2\theta\right) + 2xy\cos\theta\sin\theta - 2xy\cos\theta\sin\theta + x^2\left(\cos^2\theta+\sin^2\theta\right)$$

Hence, we use trig identities to conclude that

$$(x')^2+(y')^2 = x^2+y^2$$

which implies that a general vector's length is preserved under this transformation. Now we want to show that dot products are preserved. Let $\mathbf{a}=(a^1,a^2)$ and let $\mathbf{b}=(b^1,b^2)$. Under the transformation defined by Eq. 1, we see that

$$a'^1 = a^1\cos\theta + a^2\sin\theta$$

$$a'^2 = -a^1\sin\theta + a^2\cos\theta$$

$$b'^1 = b^1\cos\theta + b^2\sin\theta$$

$$b'^2 = -b^1\sin\theta + b^2\cos\theta$$

Then we have that, after some rearranging,

$$a'^1b'^1 + a'^2b'^2 = a^1b^1\left(\cos^2\theta+\sin^2\theta\right) + a^2b^2\left(\sin^2\theta+\cos^2\theta\right) = a^1b^1+a^2b^2$$

and thus, the dot product is preserved.

2.6. Repeat questions 2.1-2.5 for the case of an *active* transformation, where $\mathbf{v}_1$ and $\mathbf{v}_2$ are rotated counterclockwise by θ to become vectors $\mathbf{v}_1'$ and $\mathbf{v}_2'$. In doing so, you should determine the matrix R' such that $\mathbf{v}_1' = R'\mathbf{v}_1$ and $\mathbf{v}_2' = R'\mathbf{v}_2$. This exercise shows that active and passive transformations are *inverses* of each other: rotating a vector with respect to a fixed coordinate

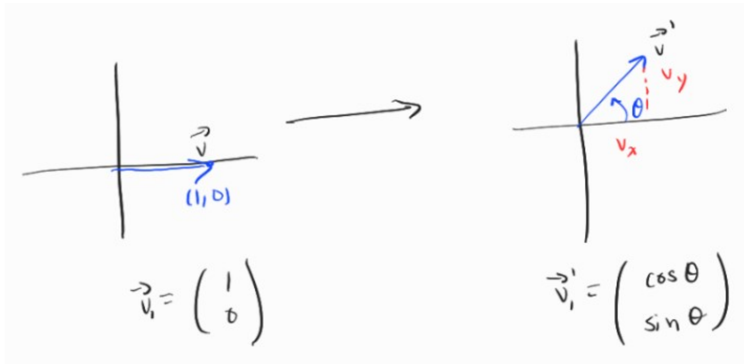

Figure 4: Active transformation that rotates the basis vector (1, 0) counterclockwise by some angle.

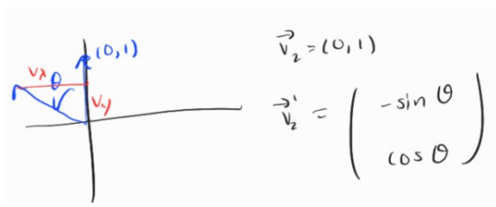

Figure 5: Active transformation that rotates the basis vector (0, 1) counterclockwise by some angle.

system is the same thing as rotating the coordinate system with respect to a fixed vector, but in the opposite direction. This is a great example of relativity!

Let's first look at an active transformation that rotates the basis vector $\mathbf{v}_1 = (1, 0)$ and leaves the coordinate system stationary. In Fig. 4, we see that this transformation sends (1, 0) to $(\cos\theta, \sin\theta)$. Furthermore, we see in Fig. 5 that the same transformation applied to the basis vector (0, 1) sends it to $(-\sin\theta, \cos\theta)$. So, we can conclude that our active transformation is described by the matrix

$$R' = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

We can calculate the inverse of the passive transformation defined earlier to conclude that this active transformation is its inverse. By performing similar calculations to part 2.5, it's trivial to show that this transformation also preserves vector lengths and dot products.

From your work in Part 2, we can see that one way to define a vector in two dimensions is as a two-component quantity (x, y) whose components after a change of coordinates are related to the old components by multiplication by a rotation matrix. In fact, this definition is perfectly valid for vectors in three dimensions, as long as we know how to define a rotation in three dimensions (we will spend a good deal of time on this later in the semester). You might be wondering why we are only dealing with rotation matrices - aren't there other kinds of coordinate transformations as well? It turns out that the only length- and angle-preserving transformations (in any dimension) are rotations, translations, and flips. You will see an example of translations in Part 3 below.[1]

[1] Flips are more complicated, because they change the orientation of a coordinate system, which turns the right-

You should also beware of the difference between active and passive transformations. Typically in physics, we deal with active transformations, but you can figure out which one you need to use by thinking carefully about the physical situation at hand, as we will see in the rest of this discussion.

Part 3: Galilean transformations as matrices

Recall the Galilean transformations from Discussion 1:

$$\Delta t = \Delta t' \tag{3}$$
$$\Delta x = \Delta x' + v\Delta t' \tag{4}$$

These are *linear transformations* because the coordinate distances Δx and Δt in frame S are linear combinations of the coordinates $\Delta x'$ and $\Delta t'$ in the moving frame S'. This means we can write the Galilean transformations as matrix equations. If we synchronize the origins of the two coordinate systems, we can drop the Δ's; this is equivalent to treating $(\Delta t, \Delta x)$ as a vector $\mathbf{x}$ which doesn't have a preferred origin. However, it is not the same kind of vector we saw in Part 1, because it combines time and space coordinates! This is a warm-up exercise for understanding *spacetime vectors* which are the essential ingredients of special relativity. But for this part, assume all velocities are small so that Galilean relativity applies.

3.1. Letting $\mathbf{x} = (t, x)$ and $\mathbf{x}' = (t', x')$, write Eqs. (3)–(4) (without the Δ's) in the form $\mathbf{x} = M\mathbf{x}'$, and find the matrix M. Does M preserve the "length" $\sqrt{t^2 + x^2}$ of the vector $\mathbf{x}$?

$$t = t'$$

$$x = x' + vt'$$

$$\begin{pmatrix} t \\ x \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ v & 1 \end{pmatrix} \begin{pmatrix} t' \\ x' \end{pmatrix}$$

Hence,

$$M = \begin{pmatrix} 1 & 0 \\ v & 1 \end{pmatrix}$$

To test whether M preserves the length of the vector, we write x' in terms of x:

$$x' = x - vt$$

and then evaluate

$$(t')^2 + (x')^2 = t^2 + (x - vt')^2 = t^2 + x^2 - 2xvt + v^2t^2 \neq t^2 + x^2$$

Therefore, M does not preserve the length of the vector.

hand rule for vectors into a left-hand rule; you can try it out by flipping o ver y our t ransparency a nd p erforming the manipulations described in the questions above.

3.2. Set $x' = 0$, which describes an event located at the origin of system S'. For $t \neq 0$, does the origin of S coincide with the origin of S'? This is another difference between Galilean transformations and the rotation matrices we saw in Parts 1–2; Galilean transformations can move the spatial components of vectors away from the origin, which means they describe *translations.*[2]

Let $x' = 0$. Since $x = x' + vt' = vt'$, $x = 0$ only if $t = 0$ (since $v = 0$ means the two frames are the same frame). Hence, for $t \neq 0$, the origins do not coincide.

3.3. Find the inverse transformation matrix M' which transforms $\mathbf{x}'$ to $\mathbf{x}$. (Feel free to refer back to the results of Discussion 1.)

We see that we want the equation $x' = -vt + x$ along with $t' = t$. Comparing this to the form $\mathbf{x}' = M^{-1}\mathbf{x}$, we see that such an inverse matrix must be

$$M^{-1} = \begin{pmatrix} 1 & 0 \\ -v & 1 \end{pmatrix}$$

3.4. Imagine a train of length L at rest, the front end of which can be described by a spacetime vector $\mathbf{x} = (t, L)$. If the train suddenly starts moving with velocity v in the $+x$ direction, find its new spacetime vector $\mathbf{x}'$ by performing an *active* Galilean transformation. Show that this description is equivalent to a *passive* transformation between the moving frame S' and the stationary frame S, using the inverse Galilean transformation and exchanging the roles of $\mathbf{x}'$ and $\mathbf{x}$.

If the front of the train is described in frame S by (t, L), and then the train starts moving at speed v, the front of the train is then defined in the S frame by $\mathbf{x} = (t, L + vt)$. If we want the coordinates of the front of the train in the S' frame which is fixed to the train, we use the following transformation

$$\mathbf{x}' = M\mathbf{x} = \begin{pmatrix} 1 & 0 \\ -v & 1 \end{pmatrix} \begin{pmatrix} t \\ L + vt \end{pmatrix} = \begin{pmatrix} t \\ L \end{pmatrix}$$

Hence, we see that the S' frame is fixed to the train, since this is the same vector that described the stationary train in S.

Part 4: Vectors in Newtonian mechanics

4.1. In terms of the definitions and concepts developed in Parts 1–2, explain in a sentence or two what it means to say that force is a vector quantity. Contrast this with the hypothetical case of a force "vector" whose components are *constant* in *all* coordinate systems, and make sure you understand this difference. Give a real-world application – for example, what do those two contrasting situations mean when you do an experiment standing upright (one coordinate system) versus standing on your head (another coordinate system related by a rotation)?

Force being a vector quantity means that its exact components depend upon the coordinate system choice being used the describe it, in contrast with the hypothetical force "vector"

[2] For the mathematically curious, Galilean transformations are examples of *affine tranformations*.

whose components are independent of the coordinate system choice. From the practical perspective, noticing that an object moves in different directions if pushed on in different directions hints at force being a vector quantity. Additionally, noticing that you can arrange more than two non-collinear forces to cancel each other out cements the concept of force as a vector. In a hypothetical experiment seen by two observers whose coordinate systems are flipped by 180 degrees relative to each other, both observers will have different descriptions of the force components involved, but their description of the motion of the object will agree as long as the force vector transforms with the coordinate system.

4.2. Explain why Newton's Second Law, $\mathbf{F} = m\mathbf{a}$, is equally valid in two coordinate systems related by a rotation. This means that we are free to choose the most convenient coordinate system to solve a particular problem, and the physics will remain unchanged, *provided* we remember to transform the components of the various vectors according to the rules laid out above. One way of interpreting this result is that, if Newton's Second Law holds no matter the orientation of the coordinate axes, then the laws of physics are completely *symmetric* with respect to rotations. Also, combined with the results of Discussion 1 and Part 3 above, this assures the invariance of the Second Law for the most general inertial systems (rotations and/or translations at constant velocity). All of these considerations will remain true in the context of special relativity.

In a coordinate system related by a simple rotation the components of the force are shifted by the exact same transformation that will shift the the components of the acceleration. That is to say, a rotation cannot ever make a force disappear in the new frame and likewise neither can it make an acceleration disappear. So if acceleration is preserved under Galilean transformations (including rotations), then we must conclude that forces are preserved as well, allowing for Newton's second law to hold in all coordinate systems related by constant velocity motion, translation, or rotation.

4.3. Explain why Newton's Third Law (for every action there is an equal and opposite reaction) holds in all coordinate systems, by using the fact that rotations preserve angles.

Since rotations are angle-preserving, the angle of the force vector relative to the object it acts upon will also be preserved, seeing as the coordinate of the object will rotate with the force. As a result, the "equal and opposite" force that arises as a result of Newton's third law will still act in the same opposite collinear direction as before. Thus, Newton's third law will hold in all coordinate systems.